A chiral Lagrangian for weak single-pion production: Lagrangian and form-factor substitution rules

Abstract

This note documents the model implemented in `ChiPT.fr`: a chiral Lagrangian with nucleons and the $\Delta(1232)$, gauged so that the electromagnetic and weak currents are generated by the same covariant derivative, and exported through FeynRules to a UFO model for use in an automated matrix-element generator. We give the Lagrangian, then the rules by which momentum-dependent form factors are attached to the generated vertices, and finally the symmetry checks that fix or calibrate every normalisation in the weak sector. The emphasis throughout is on what is *predicted* by the gauging and what has to be supplied by hand.

1 Scope and conventions

The model describes $\gamma N \to \pi N$ and the charged- and neutral-current processes $\nu N \to \ell\, N'\pi$ at and below the $\Delta(1232)$ resonance. Fields are the pion triplet π^a, the nucleon doublet $N = (p, n)^{\mathsf{T}}$, and the Rarita–Schwinger field Δ_i^μ. We use $u = \exp(i\,\pi^a\tau^a/2f_\pi)$, $U = u^2$, and the metric $g_{\mu\nu} = \mathrm{diag}(+,-,-,-)$.

External left- and right-handed sources carry the gauge fields,

$$l_\mu = \frac{g_2}{2}\Big[V_{ud}\sum_{i=1,2}\tau^i W_\mu^i + \tau^3 W_\mu^3\Big] + \frac{g_1}{2}\,B_\mu, \tag{1}$$

$$r_\mu = \frac{g_1}{2}\big(\tau^3 + \mathbf{1}\big)B_\mu, \tag{2}$$

so that the CKM factor appears only on the charged components, as it must: the neutral current and the photon are not Cabibbo suppressed.

The chiral vielbein and connection follow from these in the usual way,

$$u_\mu = i\Big[u^\dagger\big(\partial_\mu - ir_\mu\big)u - u\big(\partial_\mu - il_\mu\big)u^\dagger\Big], \tag{3}$$

$$\Gamma_\mu = \tfrac{1}{2}\Big[u^\dagger\big(\partial_\mu - ir_\mu\big)u + u\big(\partial_\mu - il_\mu\big)u^\dagger\Big]. \tag{4}$$

Writing $u_\mu = u_\mu^a\,\tau^a$, it is useful to split the isovector projection u_μ^a into the piece that survives when the pion field is switched off and the remainder,

$$u_\mu^a\Big|_{\text{source}} \equiv \lim_{f_\pi\to\infty} u_\mu^a, \qquad u_\mu^a\Big|_{\pi} \equiv u_\mu^a - u_\mu^a\Big|_{\text{source}}. \tag{5}$$

In the code these are `ChiVielbeinVWeak` and `ChiVielbeinVPion`. The split is exact and carries no approximation; it exists so that the two halves can be given different form factors, since the pion vertex and the weak transition they generate are measured in different experiments.

Which source pairs with which chiral factor. Equation (3) pairs r_μ with $u^\dagger\cdots u$ and l_μ with $u\cdots u^\dagger$. The opposite pairing,

$$\tilde{u}_\mu = i\Big[u^\dagger\big(\partial_\mu - il_\mu\big)u - u\big(\partial_\mu - ir_\mu\big)u^\dagger\Big], \tag{6}$$

has the *same* pion part — the two differ only in the external-source half, and only through $l_\mu - r_\mu$, so for a purely electromagnetic source they are identical. They are not identical for the W, which lives in l_μ alone: at $u = 1$ the source halves of u_μ and $\tilde{u}_\mu$ are $r_\mu - l_\mu$ and $l_\mu - r_\mu$. Since the axial source is $a_\mu = (r_\mu - l_\mu)/2$ and the W is purely left-handed, only one of the two reproduces the Standard Model's $V - A$ structure once the sign of the pion coupling is fixed. The nucleon sector uses $\tilde{u}_\mu$ throughout (`LMatTilda`, `RMatTilda`), and the Δ sector must therefore use it as well; the model does this by taking $-u^a_\mu|_{\text{source}}$ wherever the source half enters the Δ Lagrangian, which is the same thing through $\mathcal{O}(\pi)$. Section 4 gives the convention-free test that enforces it.

The Δ is parametrised at the pole, throughout. Four numbers describe the resonance — its mass, the width in its propagator, $\Gamma(\Delta \to N\pi)$, which since Sec. 4 *defines* h_A, and $\Gamma(\Delta \to N\gamma)$, which fixes G_1 and G_2 — and all four are pole quantities. The mass and propagator width are the PDG pole position, $M_\Delta = 1.209$ GeV and $\Gamma_\Delta = 0.102$ GeV; the two partial widths are the Breit–Wigner numbers scaled to that total at fixed branching ratio,

$$f = \frac{0.102}{0.117 + 0.00066} = 0.866905, \qquad \Gamma(N\pi) = 0.101428, \qquad \Gamma(N\gamma) = 0.000572 \text{ GeV}, \tag{7}$$

preserving $\text{BR}(N\gamma) = 0.5609\%$ exactly. Earlier versions of this model mixed the two parametrisations, taking the mass and propagator width at the pole while reading the partial widths at the Breit–Wigner mass 1.232 GeV, and since $h_A \propto \sqrt{\Gamma(N\pi)}$ evaluated at M_Δ the mismatch propagated into every coupling derived from a width.

Three things follow, and none of them was the reason for making the change, which is what makes them worth recording:

1. The partial widths now *sum* to the propagator width, so the resonance's elasticity is 1 by construction. Previously the vertices delivered 117.66 MeV through a propagator built for 102, a 15% mismatch, and the $\pi^+ p$ elastic cross section overshot the unitarity bound by a factor 2 at the peak (Sec. 4.1).

2. The UFO's width parameter and the `DECAY` block of the generated card now carry the same number, so the analytic engine and the generator no longer disagree by 17% on the same quantity.

3. Photoproduction, which constrains the *product* $G_i h_A$ and which the width parametrisation therefore had no obligation to respect, moves into agreement with the semi-analytic reference of Fig. 33 of nucl-th/0509020v2. Peak cross sections from the same UFO, evaluated with the analytic engine:

peak σ / reference	$\gamma p \to p\pi^0$	$\gamma p \to n\pi^+$	$\gamma n \to n\pi^0$	$\gamma n \to p\pi^-$
mixed parametrisation	1.36	1.24	1.37	1.18
pole parametrisation	1.02	1.01	1.03	1.02

Through the generator the same comparison gives 1.14, 1.05, 1.14 and 1.03; the residual on the π^0 channels is the $\sqrt{X_\pi}$ placement discussed in Sec. 4.2, which the analytic engine applies at amplitude level and the vertex form factor cannot. The mixed parametrisation was what made the measured widths and the photoproduction fit look incompatible — the product $G_i h_A$ was off by 1.20, and scaling both couplings by $\sqrt{f} = 0.931$ brings it to 1.04.

2 The Lagrangian

The total Lagrangian assembled in `LChiPT` is

$$\mathcal{L} = \mathcal{L}_\pi + \mathcal{L}_N + \mathcal{L}_{NNV} + \mathcal{L}_{NNA} + \mathcal{L}_\Delta + \mathcal{L}_{\pi N\Delta} + \mathcal{L}_{VN\Delta} + \mathcal{L}_{\mathrm{FF}}, \tag{8}$$

where $\mathcal{L}_{\mathrm{FF}}$ collects the form-factor terms of Sec. 3.

Mesons and nucleons. $\mathcal{L}_\pi$ is the order-p^2 chiral Lagrangian, $\frac{f_\pi^2}{4}\mathrm{Tr}\big[D_\mu U (D^\mu U)^\dagger\big]$ with $D_\mu U = \partial_\mu U - i r_\mu U + i U l_\mu$, expanded to the order needed. The nucleon sector is

$$\mathcal{L}_N + \mathcal{L}_{NNV} + \mathcal{L}_{NNA} = \bar N\big(i\not{D} - M_N\big)N + \bar N \not{V} N - g_A\, \bar N \not{u}\,\gamma_5 N, \tag{9}$$

with the vector coupling supplied by the chiral connection and the axial one by the vielbein. Because both descend from Eqs. (1)–(2), the photon, W and Z couplings to the nucleon are not independent inputs.

The $\Delta(1232)$. $\mathcal{L}_\Delta$ is the Rarita–Schwinger kinetic and mass Lagrangian with off-shell parameter A_Δ, chirally covariantised through the same connection.

The gauged $\pi N\Delta$ coupling. The $\pi N\Delta$ interaction is taken in the gauge-invariant (Pascalutsa) form, built from the vielbein rather than from $\partial_\mu\pi$,

$$\mathcal{L}_{\pi N\Delta} = -\frac{h_A}{M_\Delta}\,\epsilon^{\mu\nu\lambda\sigma}\,\partial_\mu \bar\Delta^i_\nu\,\gamma_\sigma\gamma_5\left[u^i_\lambda\Big|_\pi - u^i_\lambda\Big|_{\mathrm{source}}\right] N \;+\; \mathrm{h.c.} \tag{10}$$

This is the central structural point of the model. Because u_λ contains both $\partial_\lambda\pi$ and the external sources, Eq. (10) generates *simultaneously* the strong $\Delta \to N\pi$ vertex and the axial $N \to \Delta$ transition. Using the split of Eq. (5), the first comes from $u_\lambda|_\pi$ and the second from $u_\lambda|_{\mathrm{source}}$. The axial transition strength C_5^A is therefore *predicted* in terms of h_A rather than fitted; with the couplings used here the Adler decomposition of Sec. 4 returns $C_5^A(0) = 1.271$, against the phenomenological 1.2. It inherits the width parametrisation of Sec. 1 through h_A, and would read 1.365 on the mixed convention this model used previously.

The relative minus between the two halves is the $\tilde u_\mu$ pairing of Eq. (6), written out. It is not a phase convention: the Δ and W field phases cancel from the ratio of the two couplings this single term generates, and that ratio must match the nucleon's, which is built from the same $\tilde u_\mu$. Writing u_λ here instead of the bracket leaves the strong vertex untouched and flips the whole weak $N \to \Delta$ transition against the Born terms.

The vector $N \to \Delta$ transition. The electromagnetic $N \to \Delta$ vertex is the standard two-term form with couplings G_1 (magnetic, dual field strength) and G_2 (electric). To obtain its weak partner without introducing new parameters, the photon field strength is replaced by the field strength of the *isovector source*,

$$v^a_\mu = \begin{cases} \dfrac{V_{ud}}{2s_W}\,W^a_\mu, & a = 1,2, \\[2mm] \dfrac{1}{2s_W}W^3_\mu + \dfrac{1}{2c_W}B_\mu, & a = 3, \end{cases} \tag{11}$$

$$v^a_{\mu\nu} = \partial_\mu v^a_\nu - \partial_\nu v^a_\mu, \tag{12}$$

$$\mathcal{L}_{VN\Delta} = \kappa \sum_a \bar N\Big[iG_1\,\tilde v^a_{\mu\nu} + G_2\,\gamma_5\,v^a_{\mu\nu}\Big]\partial^\mu \Delta^{\nu,a} + \mathrm{h.c.} \tag{13}$$

Equation (11) is normalised so that its third component carries the photon with coefficient exactly 1; the electromagnetic vertex is therefore reproduced unchanged, and the W and Z transitions follow with no new input.

Equation (11) is $(2/e)\,v_\mu$ with $v_\mu = (l_\mu + r_\mu)/2$ read off Eqs. (1)–(2), and nothing in it is chosen. In particular the charged components carry the same sign as in the nucleon vector current, which uses the same v_μ: both give W^1_μ the coefficient $+g_2 V_{ud}/4$. A relative minus between the two would make one isovector source appear with opposite charged components in the two sectors, which is not an isospin rotation of the model but of half of it.

Subleading axial $N \to \Delta$ structures. Adler's C_4^A and C_6^A are added as separate operators built from the same source projection $-u_\mu|_{\text{source}}$ — the same sign as in Eq. (10), so that all three axial structures move together — so that all isospin factors and V_{ud} come along automatically:

$$\mathcal{L}_{C_4^A} = C_4^{A,N}\Big[g^{\alpha\lambda}\,\partial_\nu \bar{\Delta}^{\,i}_{\alpha}\,\partial^\nu u^i_\lambda\Big|_{\text{source}} - \partial^\lambda \bar{\Delta}^{\,i}_{\alpha}\,\partial^\alpha u^i_\lambda\Big|_{\text{source}}\Big] N + \text{h.c.}, \tag{14}$$

$$\mathcal{L}_{C_6^A} = -\,C_6^{A,N}\,\partial_\alpha\Big(\partial^\lambda u^i_\lambda\Big|_{\text{source}}\Big)\,\bar{\Delta}^{\alpha,i} N + \text{h.c.} \tag{15}$$

The two normalisations are external parameters. $C_4^{A,N}$ is calibrated so that the extracted ratio is Adler's $C_4^A = -C_5^A/4$; $C_6^{A,N}$ is calibrated against PCAC as described in Sec. 4.

Induced pseudoscalar on the nucleon. The fourth Lorentz structure of the nucleon weak vertex, $q^\mu\gamma_5$, has no point-like piece in the chiral Lagrangian and is added explicitly. Since q^μ is the momentum of the gauge boson, the operator is the *divergence* of the gauge field,

$$\mathcal{L}_{PP} = -\,i\,\frac{e\,V_{ud}}{2 s_W \sqrt{2}}\,(-g_A)\,(2M_N)\,F_P(Q^2)\,\partial_\mu W^\mu\,\bar{p}\,\gamma_5\,n \;+\; \text{h.c.} \;+\; \big(Z \text{ terms}\big). \tag{16}$$

The explicit i is required by hermiticity, since $\bar{\psi}\gamma_5\psi$ is anti-hermitian while $\bar{\psi}\gamma^\mu\gamma_5\psi$ is hermitian; the overall sign is fixed by the PCAC test. Because q^μ contracted with the leptonic tensor is proportional to m_ℓ^2, this term contributes to rates only for a massive charged lepton.

3 Form-factor substitution rules

The chiral Lagrangian of Sec. 2 generates *point-like* vertices: the nucleon couples to the photon with charge 1 and to the axial current with g_A, and nothing depends on Q^2. Turning that into a usable model means attaching momentum dependence to each vertex without disturbing the gauging that produced it. This section gives those substitutions operator by operator.

3.1 The three patterns

Every substitution used here is one of three kinds.

(a) Additive correction to a point-like vertex. Where the chiral Lagrangian already supplies the operator, the term *stays*, and a correction is added beside it:

$$\mathcal{O}_{\text{point}} \;\longrightarrow\; \mathcal{O}_{\text{point}} + \big(F(Q^2) - 1\big)\,\mathcal{O}_{\text{point}} \;=\; F(Q^2)\,\mathcal{O}_{\text{point}}. \tag{17}$$

The result is the substitution one wants, $1 \to F(Q^2)$, but it is written as an addition rather than a replacement. This matters: the point-like coupling is not a free-standing term but a piece of $\bar{N}\not{V}N$ or $-g_A\bar{N}\not{A}\gamma_5 N$, which also carry the pion couplings that chiral symmetry ties to it. Multiplying those Lagrangian terms by F would attach the nucleon form factor to the pion vertices as well. Writing the correction separately touches only the vertex intended. Since $F(0) = 1$ by construction for this class, the correction vanishes at the photon point and no existing fit is disturbed.

(b) New operator with no point-like counterpart. Some structures are simply absent from the chiral Lagrangian at this order — the Pauli term, the induced pseudoscalar, Adler's C_4^A and C_6^A. There is nothing to correct, so the operator is added with a *bare* form factor carrying its own absolute normalisation:

$$0 \;\longrightarrow\; F(Q^2)\,\mathcal{O}_{\rm new}. \tag{18}$$

These F are not normalised to 1 at $Q^2 = 0$. The neutron Dirac form factor belongs to this class rather than to (a), because the point-like neutron charge is zero: F_1^n multiplies $\bar{n}\not{A}n$ bare, not as $(F_1^n - 1)$.

(c) Replacement of a whole term. Where the point-like term is removed from $\mathcal{L}$ altogether and rewritten, the form factor multiplies the replacement directly. This is used for the two Δ couplings, whose form factors are attached at different places within one term.

3.2 The substitutions, operator by operator

Nucleon electromagnetic current. Pattern (a) for the proton Dirac coupling, (b) for the neutron and for both Pauli terms:

$$e\,\bar{p}\not{A}p \;\longrightarrow\; eF_1^p\,\bar{p}\not{A}p + eF_1^n\,\bar{n}\not{A}n + \frac{e}{4M_N}F^{\mu\nu}[F_2^p\,\bar{p}\sigma_{\mu\nu}p + F_2^n\,\bar{n}\sigma_{\mu\nu}n]\,. \tag{19}$$

Nucleon charged current. Pattern (a) on the Dirac and axial couplings, (b) on weak magnetism and on the induced pseudoscalar:

$$\frac{eV_{ud}}{2s_W\sqrt{2}}\,\bar{p}\big[\not{W} - g_A\not{W}\gamma_5\big]n \;\longrightarrow\; \frac{eV_{ud}}{2s_W\sqrt{2}}\,\bar{p}\Big[F_1^V\not{W} - g_A F_A\not{W}\gamma_5 + \frac{F_2^V}{4M_N}\,W^{\mu\nu}\sigma_{\mu\nu} - i\,g_A\,2M_N F_P\,(\partial_\mu W^\mu)\,\gamma_5\Big]n, \tag{20}$$

with $F_1^V = F_1^p - F_1^n$ and $F_2^V = F_2^p - F_2^n$ by CVC. The neutral current follows the same pattern with the charge combinations of Table 1.

Pion-in-flight and contact term. Once F_1^V sits on the nucleon poles, CVC requires it on the rest of the vector background too. Both are pattern (a). The substitution is made on the *sources* rather than on the bosons, so that one operator dresses the photon, the Z and the W at once; the different weights are already in the model's own couplings.

Which Dirac structure of the contact term belongs to which current is the one thing here that is easy to get backwards, and is worth stating explicitly. Writing $u = \exp(i\Phi)$ with $\Phi = \boldsymbol{\pi}\cdot\boldsymbol{\tau}/2f_\pi$,

$$\tfrac{1}{2}(\tilde{l} + \tilde{r}) = v - i[\Phi, a] + \mathcal{O}(f_\pi^{-2}), \qquad \tfrac{1}{2}(\tilde{l} - \tilde{r}) = a - i[\Phi, v] + \mathcal{O}(f_\pi^{-2}), \tag{21}$$

so the *one-pion* piece of each combination carries the *other* source. The contact term generated by $\mathcal{L}_{NN}^V$ therefore has the γ^μ (Weinberg–Tomozawa) structure but multiplies the axial source, and belongs to $\langle\pi N|A^\mu|N\rangle$; the one generated by $\mathcal{L}_{NN}^A$ has the $\gamma^\mu\gamma_5$ (Kroll–Ruderman) structure but multiplies the vector source, and is the piece CVC constrains:

$$\mathcal{L}_{PF} \longrightarrow F_1^V\,\mathcal{L}_{PF}, \tag{22}$$

$$-g_A\,\bar{N}\not{l}\,\gamma_5 N \longrightarrow -g_A F_1^V\,\bar{N}\not{l}\,\gamma_5 N \qquad \text{(Kroll–Ruderman, vector current)}, \tag{23}$$

$$\bar{N}\not{l}\,N \longrightarrow F_\rho\,\bar{N}\not{l}\,N, \qquad F_\rho = \frac{m_\rho^2}{m_\rho^2 + Q^2} \qquad \text{(Weinberg–Tomozawa, axial current)}. \tag{24}$$

The photon settles the assignment on its own and without reference to any convention: for the photon $l = r$, hence $a = 0$, hence by Eq. (21) there is *no* γ^μ photon contact term at all —

and the UFO indeed contains none. The only photon contact term is the Kroll–Ruderman one, which is precisely the term the low-energy theorem for charged-pion photoproduction needs in order to close the electromagnetic Ward identity, so it cannot be anything but vector-current and cannot carry F_ρ. Section 4 verifies the result directly.

The neutral current needs one extra constant. $F_{Z1}^{p,n}$ do two jobs at once (Table 1): they supply the Q^2 dependence and they move the nucleon's neutral-current vector charge from the model's tree-level $\sin^2\theta_W = 1 - M_W^2/M_Z^2$ to the low-Q value, because $\frac{1}{2} - 2s_W^2$ sits near a zero. Conservation of the NC vector current needs the nucleon's isovector piece and the pion side to carry the same function,

$$g_p F_{Z1}^p - g_n F_{Z1}^n = F_1^V \,(1 - 2s_W^2\big|_{\text{low-}Q}), \tag{25}$$

whereas the model's own $Z\pi\pi$ coupling was built with the tree-level angle. The two differ by a constant 0.908 at the default values. The Z half of the background therefore carries

$$F_{Z1}^\pi(Q^2) = F_1^V(Q^2)\,\frac{1 - 2s_W^2\big|_{\text{low-}Q}}{1 - 2s_W^2\big|_{\text{tree}}}, \tag{26}$$

built the same way F_{Z1}^p is — physical combination over the model's own point-like coupling. Only the vector-current halves take it: the $Z\pi\pi$ vertex and the Kroll–Ruderman half of the Z contact term. The NC axial current is pure isovector and carries no $\sin^2\theta_W$ at all, so the Weinberg–Tomozawa half keeps F_ρ. The photon needs no analogue — its charges are 1 and 0 with no mixing angle in them — and neither does the W.

The $\pi N\Delta$ coupling. Pattern (c). Equation (10) is replaced, and the split of Eq. (5) lets the two vertices it generates carry different form factors:

$$u_\lambda^i \;\longrightarrow\; F_{\pi N\Delta}(P_\Delta^2)\; u_\lambda^i\Big|_\pi \;+\; F_{C_5^A}(Q^2)\; u_\lambda^i\Big|_{\text{source}}. \tag{27}$$

This is the substitution that most needs the split. The strong $\Delta \to N\pi$ vertex is damped by the resonance form factor $\sqrt{X_\pi}$, a function of the Δ leg's invariant mass; the weak $N \to \Delta$ transition is damped by the axial dipole, a function of Q^2. They are different functions of different invariants sitting in one Lagrangian term, and only Eq. (27) separates them. Note that exactly one factor of $\sqrt{X_\pi}$ appears per Δ diagram, on the strong vertex, so it multiplies the vector and axial transitions alike.

The vector $N \to \Delta$ transition. Pattern (c):

$$\mathcal{L}_{VN\Delta} \;\longrightarrow\; F_{VN\Delta}(Q^2)\,\mathcal{L}_{VN\Delta}, \qquad F_{VN\Delta} = \frac{1}{(1 + Q^2/M_V^2)^2\,(1 + Q^2/4M_V^2)}. \tag{28}$$

The photon vertex is covered by the same symbol, and since $F_{VN\Delta}(0) = 1$ real photoproduction is untouched.

C_4^A and C_6^A. Pattern (b), with $F_{C_5^A}$ riding along on C_4^A so that it keeps exactly C_5^A's Q^2 dependence, as $C_4^A = -C_5^A/4$ requires, and $F_{C_6^A} = F_{C_5^A}/(Q^2 + m_\pi^2)$ on C_6^A.

3.3 Implementation constraints

Three properties of the FeynRules interface govern how the substitutions above have to be written. A form factor is declared in `M$FormFactors` as

`Particles → {...}, Value → "FW1(-P(-9,3)**2)", ParameterName → FFW1A`

and used in the Lagrangian as an ordinary symbol; the generator evaluates the named function at run time through a plugin (`FormFactors.C`), so the parameterisation can be changed without regenerating the model.

1. **The leg index is spliced verbatim.** `P(-9,3)` refers to leg 3 *of the generated vertex* and is not relabelled to match the exported ordering. The argument is $-P_3^2 = Q^2$ when leg 3 is the gauge boson, but $+P_2^2 = W^2$ when leg 2 is the Δ — both occur, in Eq. (27). A mismatch is silent, so the export script checks *every* placement — 160 of them in the current build — against the spin of the leg it names: $-P^2$ must name a gauge boson and $+P^2$ must not. `tools/ff_audit.py` runs the same test on an already-built UFO.

2. **One form factor per Lorentz structure.** A structure carrying a product fails at generation time. This is a genuine constraint rather than a convention: the product in Eq. (27) would be $F_{C_5^A}(Q^2)F_{\pi N\Delta}(W^2)$ if both were placed on the same half, and two functions of *different* invariants cannot be collapsed into one single-argument function at all. The split of Eq. (5) is what avoids this.

3. `Series` **at infinity truncates by term count, not by power.** `Series[expr, {fPi, Infinity, n}]` keeps $n+1$ terms counting from the leading power, so `Series[a/fPi + b/fPi^2, {fPi, Infinity, 0}]` returns `a/fPi` and not 0. The counterterms here are built by splitting a source matrix into its pion-independent and one-pion pieces, and the matrix entries that hold the charged-pion contact term have no f_π^0 term at all. An order-0 truncation therefore returned the contact term itself and the subtraction that followed left zero, which is how V8 came to export the $\pi^\pm$ contact terms bare while the π^0 ones were dressed. Extracting the coefficient of $1/f_\pi$ explicitly cannot fail that way, and is what the model now does.

4. **Sums of Lagrangian terms must be assigned before they are added.** Written inline, `FeynmanRules[(a-1)*X + (b-1)*Y]` can silently return the vertices of X alone; evaluating X and Y into variables first and then summing gives all of them. This was already known for the full `LChiPT` sum and bites again for the two-term pion-in-flight counterterm, where the entire Z half disappears.

5. **Structures are grouped by coupling.** FeynRules merges terms sharing a coupling constant into one Lorentz structure, so two operators with identical prefactors combine and, if both carry form factors, violate point 2. The remedy is a distinct constant on the new operator: the $2M_N$ in Eq. (16) and the $C_6^{A,N}$ in Eq. (15) serve this purpose as well as their physical one, exactly as $1/4M_N$ does for the Pauli term.

3.4 The form factors

Table 1 lists every form factor, its argument and its definition. Ratios are normalised to 1 at $Q^2 = 0$ and multiply a coupling the model already possesses, so they change nothing at the photon point. Absolute form factors multiply operators that have no point-like counterpart.

Table 1: Form factors, their arguments and definitions. Λ-type constants live in the `CHIPTFF` block of the parameter card.

name	arg.	definition
Electromagnetic nucleon		

name	arg.	definition
`F1p`,`F1n`	Q^2	Dirac, from the Kelly (or Arrington–Hill) Sachs parameterisation
`F2p`,`F2n`	Q^2	Pauli, likewise; $F_2^p(0) = \kappa_p$, $F_2^n(0) = \kappa_n$
Charged current, nucleon		
`FW1`	Q^2	$F_1^p - F_1^n$ (CVC; ratio, $= 1$ at $Q^2 = 0$)
`FW2`	Q^2	$F_2^p - F_2^n$ (CVC; absolute)
`FWA`	Q^2	$F_A(Q^2)/F_A(0)$, axial z-expansion
`FPP`	Q^2	$F_A(Q^2)/F_A(0)/(Q^2 + m_{\pi^\pm}^2)$; the $2M_N$ of $F_P = 2M_N F_A/(Q^2 + m_\pi^2)$ is carried by the coupling
Neutral current, nucleon		
`FZ1p`,`FZ1n`	Q^2	$\left[\frac{1}{2}(F_1^p - F_1^n) - 2s_W^2 F_1^{p,n}\right]$ divided by the model's own point-like charge. `FZ1n` is 1 at $Q^2 = 0$; `FZ1p` is not, because it also carries the shift of $\sin^2\theta_W$ from the on-shell value to the $Q \to 0$ one
`FZ2p`,`FZ2n`	Q^2	the same combination of F_2; absolute
`FZA`	Q^2	$=$ `FWA`; the NC axial current is pure isovector
`FPPZ`	Q^2	as `FPP` with m_{π^0}
`FZ1pi`	Q^2	`FW1` times the constant $(1 - 2s_W^2\|_{\text{low-}Q})/(1 - 2s_W^2\|_{\text{tree}}) = 0.908$; the pion side of the NC background, so that the isovector piece of the nucleon current and the $Z\pi\pi$ / Z contact vertices carry the same function
$N \to \Delta$		
`FpiNDelta`	P_Δ^2	$\sqrt{X_\pi(s)}$, the resonance damping; note the argument is the Δ leg's invariant mass squared, not Q^2
`FCA5`	Q^2	$\left[(1 + Q^2/M_A^2)^2(1 + Q^2/3M_A^2)\right]^{-1}$
`FVND`	Q^2	$\left[(1+Q^2/M_V^2)^2(1+Q^2/4M_V^2)\right]^{-1}$, the CVC vector dipole
`FC6A`	Q^2	`FCA5`$/(Q^2+m_\pi^2)$; PCAC fixes $C_6^A = C_5^A M_N^2/(Q^2 + m_\pi^2)$, and the $1/M_N^2$ is supplied by the two derivatives in Eq. (15)
`FVND`	—	the photon $N \to \Delta$ vertices carry *this same* function, not one of their own: Eq. (13) gauges the photon and the W/Z through one isovector source, so one form factor rides on all of them. Because `FVND` is a ratio, normalised to 1 at $Q^2 = 0$, the photoproduction fit for G_1, G_2 is not silently rescaled. (A separate `FgNDelta*` set exists in `ChiPT.fr` for the Δ-only build, where $\mathcal{L}_{\gamma N\Delta}$ is not replaced by Eq. (13); it is absent from the exported model used here)
Background diagrams		
`FW1`	Q^2	reused on the pion-in-flight ($\gamma\pi\pi$, $W\pi\pi$ and the γ/W multi-pion partners) and on the Kroll–Ruderman ($\gamma^\mu\gamma_5$) half of the contact term in every charge state, as CVC requires once it is on the nucleon poles

name	arg.	definition
`FZ1pi`	Q^2	the same two places for the Z; see above
`FrhoCT`	Q^2	$m_\rho^2/(m_\rho^2+Q^2)$ on the Weinberg–Tomozawa (γ^μ) half of the contact term, which is the half that belongs to the axial current. The photon has no such half

4 Symmetry checks and calibrated constants

Every normalisation in the weak sector is either fixed by a conservation law or calibrated against one. The checks are implemented in `tools/` and act on the generated vertices, not on the Lagrangian, so they test the exported model.

Relative sign of the axial source, and of the isovector one. Two ratios in the exported model are free of every field phase, and they are the sharpest checks in this section because they compare the model against itself with nothing convention-dependent left over.

The first is the weak axial source measured against the pion coupling *inside one sector*. Both come from the same term, so the pion, gauge-boson and baryon phases cancel, and the two sectors must agree:

$$\frac{g\big(\bar p\, n\, W^+\big)_{\gamma^\mu\gamma_5}}{g\big(\bar p\, n\, \pi^+\big)} \;=\; \frac{g\big(\bar p\, \Delta^{++} W^-\big)_{C_5^A}}{g\big(\bar p\, \Delta^{++}\pi^-\big)} \;=\; +\,\frac{i}{2}\, g\, f_\pi\, V_{ud}. \tag{29}$$

This is what enforces the $\tilde u_\mu$ pairing of Eq. (6) in the Δ sector. With u_μ instead the right-hand side of the second ratio is $-\frac{i}{2} g f_\pi V_{ud}$: the same modulus, so no rate built from the Δ alone can see it.

The second is the isospin embedding. The Δ field's charged-to-neutral Clebsch is a property of its definition alone, and the model states it through its own strong vertex,

$$\frac{g\big(\bar p\, \Delta^{++}\pi^-\big)}{g\big(\bar p\, \Delta^{+}\pi^0\big)} = -1.224745 = -\sqrt{3/2}, \tag{30}$$

and the axial transition, built from the same field, gives the same. The vector transition must therefore give it too.

CVC on the vector $N \to \Delta$ transition. With the isovector source normalised as in Eq. (11), the weak vector transition must equal $-V_{ud}\sqrt{3}/(2 s_W \sqrt{2})$ times the model's own photon transition. The generated model returns -1.29517792 against an expected -1.29517792, i.e. a ratio of $+1.00000000$ with residual 5.4×10^{-17}. The sign of the Clebsch here is *not* universal — it belongs to the Δ phase convention, and Eq. (30) is the model's statement of its own. Taking the textbook $+\sqrt{3/2}$ instead is what would put a spurious minus on the charged components of Eq. (11).

The Adler decomposition. `tools/adler_ndelta.py` fits the exported $WN \to \Delta$ vertex to the eight standard structures by least squares over all four Δ helicities, both nucleon spins and all four μ — 32 complex equations for 8 unknowns. The fit is exact to $\sim 10^{-15}$, which is itself a result: the vertex really does live in the Adler basis. Dividing out the $\sqrt{3}\; p \to \Delta^{++}$ Clebsch, at $Q^2 = 0$,

$$C_3^A = 0, \quad C_4^A = -0.318, \quad C_5^A = +1.271, \quad C_6^A = +1.119, \quad C_3^V = +2.118, \quad C_4^V = -0.989, \quad C_5^V = C_6^V = 0, \tag{31}$$

and $C_4^A/C_5^A = -0.25000000$. C_3^V is worth singling out: CVC relates it to the measured $N \to \Delta$ magnetic form factor and phenomenology puts it at 2.13, which the model reproduces to 0.6% without it having been an input anywhere — on the mixed width convention of Sec. 1 it read 2.275, 7% high. These are quoted with all form factors set to 1, so the pion pole is stripped from C_6^A and the ratio $C_6^A/C_5^A = 0.880 = M_N^2/\mathrm{GeV}^2$; restoring the pole returns $C_6^A = C_5^A\, M_N^2/(Q^2 + m_\pi^2)$, which is PCAC.

Every sign here is the standard phenomenological one, and none of them was targeted — they follow from Eqs. (29)–(30), which is the strongest evidence that those two ratios are right. Reversing the axial sign of Eq. (29) flips C_4^A, C_5^A and C_6^A and leaves the vector coefficients alone; reversing the sign of the charged components of Eq. (11) flips C_3^V and C_4^V and leaves the axial ones alone. Doing both flips all six, which is an overall phase on the transition and therefore invisible in $|\Delta|^2$ — the Δ-only cross section is unchanged to every printed digit. Two compensating sign errors of exactly this kind are visible only in the Δ–Born interference and, because the second does not touch the neutral component, in the neutral current.

Conservation of the vector current in the background. The three-point checks above test the vertices whose form factors were put in deliberately. They cannot see the failure mode that actually matters for the background, which is that the nucleon poles, the pion-in-flight and the contact term have to carry the *same* F_1^V or the four diagrams stop cancelling. `tools/ward_background.py` assembles the hadronic current from the exported vertices and contracts it,

$$q_\mu \left[\,\mathrm{NP} + \mathrm{CNP} + \mathrm{PF} + \mathrm{CT}\,\right]^\mu = 0, \qquad q_\mu \left[\,\Delta_s\,\right]^\mu = q_\mu \left[\,\Delta_u\,\right]^\mu = 0, \tag{32}$$

for $\gamma N \to \pi N$ at the photon point, for $eN \to e'N\pi$ at $Q^2 = 0.1$–$2\,\mathrm{GeV}^2$, and for the vector part of the neutral and charged currents over the same range, in every charge channel. The Δ terms are conserved one at a time because the transition is built from the photon field strength, so $q_\mu F^{\mu\nu}$ vanishes identically; the Born terms are conserved only as a group, which is what makes the test sharp. Every residual is at machine precision, $\lesssim 10^{-16}$ relative to the norm of the current itself.

Two things this test found are worth recording. The first is that the γ^μ and $\gamma^\mu\gamma_5$ halves of the contact term belong to the opposite currents from the naive reading, Eq. (21); with the two form factors interchanged the residual is $\mathcal{O}(1)$ at $Q^2 = 1\,\mathrm{GeV}^2$. The second is the $\sin^2\theta_W$ mismatch on the neutral current, which shows up as a flat 2–3% residual in the charged-pion channels only and is what F_{Z1}^π removes.

Run without `--isospin-exact` the charged-current channels keep a residual of $\sim 10^{-3}$. That is not a violation: the pion-in-flight there exchanges a π^0 while the produced pion is a $\pi^\pm$, and the identity closes only up to $m_{\pi^\pm}^2 - m_{\pi^0}^2 \simeq 1.3 \times 10^{-3}\,\mathrm{GeV}^2$. Setting the two masses equal returns the residual to 10^{-16}, which is how the test separates isospin breaking from a genuine failure.

PCAC on the nucleon. Contracting q_μ into the nucleon axial vertex must give

$$q_\mu \langle N' | A^\mu | N \rangle = 2 M_N F_A \, \frac{m_\pi^2}{Q^2 + m_\pi^2}\, \bar{u}\gamma_5 u, \tag{33}$$

i.e. a result suppressed by $m_\pi^2/(Q^2 + m_\pi^2)$. The generated vertices reproduce this prediction to all printed digits on both the W and the Z. With the opposite sign for Eq. (16) the contraction is instead *enhanced* by $1 + Q^2/(Q^2 + m_\pi^2)$, so the test separates the two possibilities by two orders of magnitude at $Q^2 = 1\,\mathrm{GeV}^2$.

PCAC on the Δ, and $C_6^{A,N}$. The same contraction on the $N \to \Delta$ current fixes $C_6^{A,N}$. Two points of technique matter. The Rarita–Schwinger leg must be projected on shell before

contracting, or the unphysical spin-1/2 components dominate and the calibration returns zero with no change in the residual. And the coefficient that drives the contraction to *zero* is the chiral-limit one; the physical value is that times $Q^2/(Q^2+m_\pi^2)$. With this done, the calibration returns

$$C_6^{A,N} = 1.556910 \quad \text{at } Q^2 = 0.05,\ 0.1,\ 0.2,\ 0.5,\ 1.0,\ 2.0\ \text{GeV}^2, \quad \text{spread } 4.1\times10^{-16}, \tag{34}$$

and the residue then reproduces $m_\pi^2/(Q^2+m_\pi^2)$ exactly. That the value is constant over a factor 40 in Q^2 is the evidence that the operator structure in Eq. (15) is correct. Note that $C_6^{A,N} = h_A$: the C_5^A transition carries h_A through the gauged coupling while Eq. (15) does not, so the ratio is h_A and the calibration must be repeated if h_A is changed — as it was by the width convention of Sec. 1.

Spin-average normalisation. The Δ width used to fit h_A and $G_{1,2}$ carries the $1/(2J+1)$ average over the resonance spin, a factor 4 in the rate and 2 in the amplitude. It is absorbed by $h_A \to 2h_A$, $G_i \to G_i/2$, which leaves the product $G_i h_A$ — the only combination photoproduction constrains — exactly invariant. This is verified rather than assumed: $\gamma N \to \pi N$ is unchanged to eight digits in all four charge channels.

4.1 Pion–nucleon elastic scattering

Every check above runs through a photon or a neutrino. $\pi N \to \pi N$ uses the same strong $\pi N\Delta$ and πNN vertices and none of the electroweak machinery, and unlike photoproduction it has a ceiling that needs no fit: $\pi^+ p$ is pure $I = 3/2$, the Δ is elastic there, and a single resonant $J = 3/2$ partial wave cannot exceed

$$\sigma_{\text{el}} \;\le\; \frac{4\pi}{k^2}\,\frac{2J+1}{(2s_1+1)(2s_2+1)} \;=\; \frac{8\pi}{k(W)^2}, \tag{35}$$

about 190 mb at the Breit–Wigner mass. The bound runs with W, so the model is compared with it at each W rather than with its value at the pole.

Figure 1 shows the comparison against the PDG compilation of $\pi^\pm p$ elastic and total cross sections. Two things are visible and both are honest reporting rather than agreement.

First, **the model still exceeds the unitarity bound near the peak**. It saturates at 1.53 at $W = 1.246$ GeV, against the measured $\pi^+ p$, which touches 1.0 and no more. The pole parametrisation of Sec. 1 improved this substantially — the mixed convention reached 1.98, a factor 2 violation — but did not remove it. A tree-level resonance with no rescattering has no mechanism to respect Eq. (35), and this is the cleanest quantitative statement of what unitarisation would have to fix; it is the same omission recorded under Watson's theorem in Sec. 5.

Second, **the $\pi^- p$ panel shows the missing $I = 1/2$ sector directly**. $\pi^- p$ elastic is $\frac{1}{3}A_3 + \frac{2}{3}A_1$ in amplitude, so the $I = 3/2$ piece the model has enters its cross section with weight 1/9; that curve is the model's own Δ-only $\pi^+ p$ result rotated by isospin and is not a prediction of $\pi^- p$. The data show a second peak at $W \simeq 1.52$ GeV that the model has no field for, and carry a background between the two resonances that it cannot generate. The gap between curve and data is therefore a picture of the $D_{13}(1520)$ and the Roper, not of a normalisation error.

4.2 Where one variable has to stand in for two

A UFO form factor takes one argument, and Sherpa's reader allows one form factor per Lorentz structure. Wherever a vertex sits between two internal lines its exact form factor is a function of two invariants, and the model has to approximate. `tools/ff_audit.py` enumerates these from the exported vertices — it also checks that every form factor's argument names the leg it was

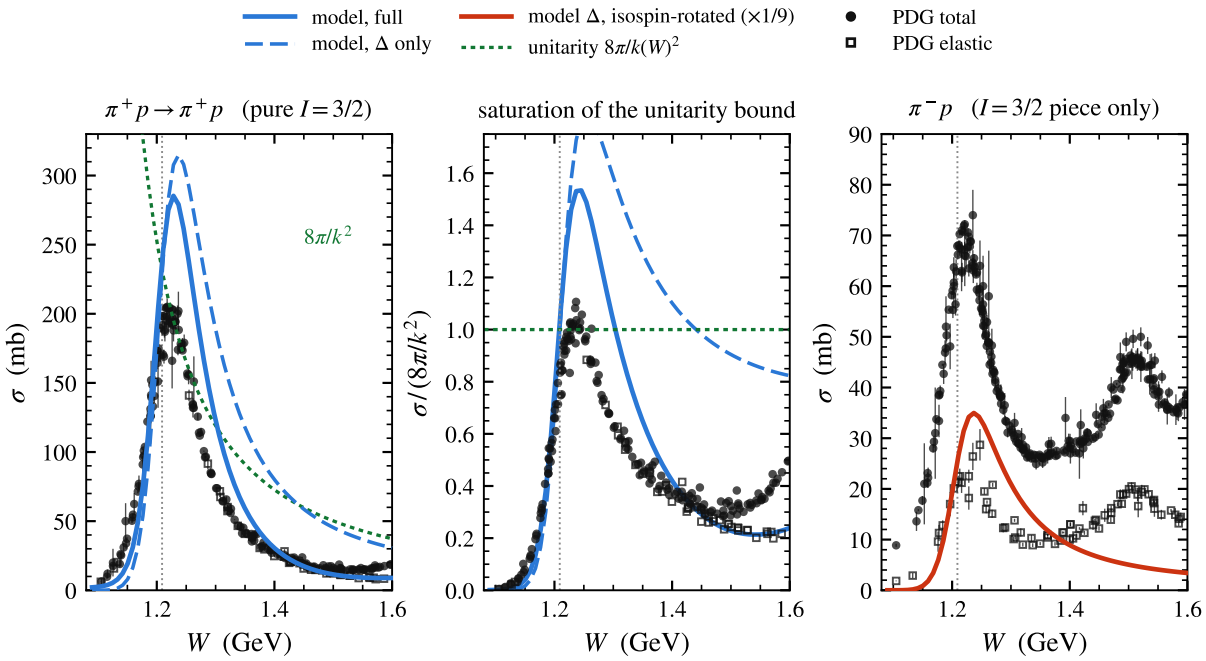


Figure 1: πN elastic scattering from the same strong vertices that build the weak Δ diagram. **Left:** π^+p, the pure $I = 3/2$ channel, with the model full and Δ-only, the unitarity bound Eq. (35), and the PDG elastic and total sets — which coincide below $W \simeq 1.35$ GeV, where the Δ is elastic. **Centre:** the same divided by the bound; the data saturate it and the model overshoots. **Right:** π^-p, with the model's Δ contribution rotated by isospin ($\times 1/9$); everything above that curve is $I = 1/2$ the model does not contain. Dotted vertical line at M_Δ. Data from the PDG hadronic cross-section compilation via `pin_data.py`.

meant to ($-P^2$ a gauge boson, $+P^2$ a hadron), which is otherwise completely silent, and lists every vertex with a hadron on it that carries no form factor at all. The cases that matter are:

- **The strong πNN vertex** is point-like in the UFO. It is internal on the nucleon leg in NP/CNP and on the pion leg in the pion-in-flight, so its exact form factor is a function of the off-shell nucleon's p^2 *and* the exchanged pion's t. This is why the Davidson–Workman $\hat{F}_B(s,u,t)$ lives at amplitude level in `fig33_from_ufo.py` and not in the model: no single-argument vertex form factor can express it.

- **The boson–$\pi\pi$ vertex** carries $F_1^V(Q^2)$ but the exchanged pion is off shell, so the exact object is $F(Q^2,t)$. Dropping the t dependence is the usual treatment and is what keeps the Ward identity exact — restoring it would require the same function on the contact term, which has no t to depend on.

- **The $N \to \Delta$ transition** depends on Q^2 and on the Δ leg's invariant mass. The model factorises the two across neighbouring vertices: $F_{C_5^A}(Q^2)$ and $F_{VN\Delta}(Q^2)$ on the transition, $\sqrt{X_\pi(W^2)}$ on the strong $\Delta \to N\pi$ vertex, one factor per Δ diagram. This is exact only if the true form factor factorises as $F(Q^2)g(W^2)$.

- **$\Delta\Delta$ vertices, the four-pion contact terms and the two-boson seagulls** are point-like. None is reached by any topology this model builds at tree level, so nothing that has been computed depends on them; they would matter for Δ radiative decay, $\pi\pi$ rescattering and Compton-like processes respectively.

5 Known limitations

- **No two-point W–π mixing.** FeynRules does not export two-point vertices, and the UFO produced here contains none. The pion pole is therefore introduced as an effective operator with the propagator absorbed into a form factor, Eq. (16). PCAC is consequently *imposed* on the nucleon and Δ currents rather than emerging from the gauging, which is a weaker statement than the CVC case. Diagonalising the mixing away is not possible within the `Definitions` mechanism, which requires a local, same-spin, non-derivative substitution.

- **Isospin breaking in the charged-current background.** The pion-in-flight in $\nu N \to \ell N \pi^{\pm}$ exchanges a π^0 while the produced pion is charged, so the vector Ward identity closes only up to $m^2_{\pi^\pm} - m^2_{\pi^0}$. The residual is $\sim 10^{-3}$ of the current. Using a common pion mass would remove it and is what the analytic comparison in `fig33_from_ufo.py` does; the UFO uses the physical masses because the generator does.

- **The Δ width is fixed, not running.** Comix takes Γ_Δ from the `DECAY` block of the parameter card and applies that one number to every propagator, s- and u-channel alike. Nothing in the UFO interface can express a width that depends on the propagator's own invariant mass, so this is not something the model can be given: it is something the generator would have to learn to do. The size of what is missing is estimated in Fig. 2, computed with the independent analytic engine `wprod_from_ufo.py` as a ratio at fixed integration grid so that the quadrature error cancels.

  Three prescriptions are compared, all normalised so that $\Gamma(M_\Delta) = \Gamma_0$ exactly, so that the comparison is of shape and nothing rescales the resonance: the k^3 law of Hernández, Nieves and Valverde, $\Gamma_0 (M_\Delta/W)(k/k_0)^3\,\Theta(W - M_N - m_\pi)$; the same with an $\ell = 1$ Blatt–Weisskopf barrier factor $F^2(q)/F^2(q_0) = (1 + (q_0 R)^2)/(1 + (qR)^2)$ at $R = 1$ fm, of which the first is the $R \to 0$ limit; and $\Gamma_0 X_\pi$ with $X_\pi = 2r^3/(1 + r^5)$, which is not a k^3 law at all — it turns over above the resonance and leaves the width between about 70 and 120 MeV over the whole $W < 1.4$ region.

  The three bracket the fixed-width answer rather than clustering around it. On $\nu_\mu p \to \mu^- p \pi^+$ between 1 and 1.5 GeV they give -16 to -20%, -9 to -11% and $+6\%$ respectively; the correction crosses unity at $E_\nu \simeq 0.7$ GeV and reaches $+60\%$ at 0.45 GeV, so it tilts the energy dependence rather than rescaling it and cannot be absorbed into C_5^A. Varying R over 0.5–1.5 fm is worth about 7%, so the first two prescriptions are ends of one soft parameter and not rival choices; the bare k^3 law reaches 563 MeV at the $W < 1.4$ cut and twenty times Γ_0 by $W = 1.8$ GeV, which is the reason not to adopt it unmodified.

  One part of this is a defect rather than a preference, and it is not visible in the figure. The crossed Δ sits at $u < 0$, always below the πN threshold, where every running form is exactly zero by the Θ function. The fixed-width run hands that propagator an imaginary part of 118 MeV that it has no business having, and no choice of Γ_0 removes it — only a width that knows its own argument does.

- **No unitarisation.** Watson's theorem is not imposed. The required phases act on πN partial-wave amplitudes and have no vertex representation; they would have to be applied to the assembled amplitude in the generator.

- **No $I = 1/2$ resonances.** The Roper $N^*(1440)$ and $D_{13}(1520)$ are absent. Both are expressible in this framework as new fields with their own transition and decay vertices.

- **No vector mesons.** ω and ρ t-channel exchange, which is required to describe $\gamma N \to \pi N$ above the Δ, is not included. By CVC the ω contribution carries over to the charged current with its coupling fixed by the photoproduction fit; the isoscalar-photon part of the ρ contribution does not, having no charged-current analogue.

- **The two-boson seagulls are not dressed.** The $VV\pi\pi$ vertices that the pion kinetic term generates are left point-like: the substitution is made on the source-linear part of $\mathrm{Tr}\,[D_\mu U\, D^\mu U^\dagger]$ only. They do not enter single-pion production with one current, so no result here depends on them, but a Compton-like calculation would need them.
- **NLO chiral counterterms** are not included.

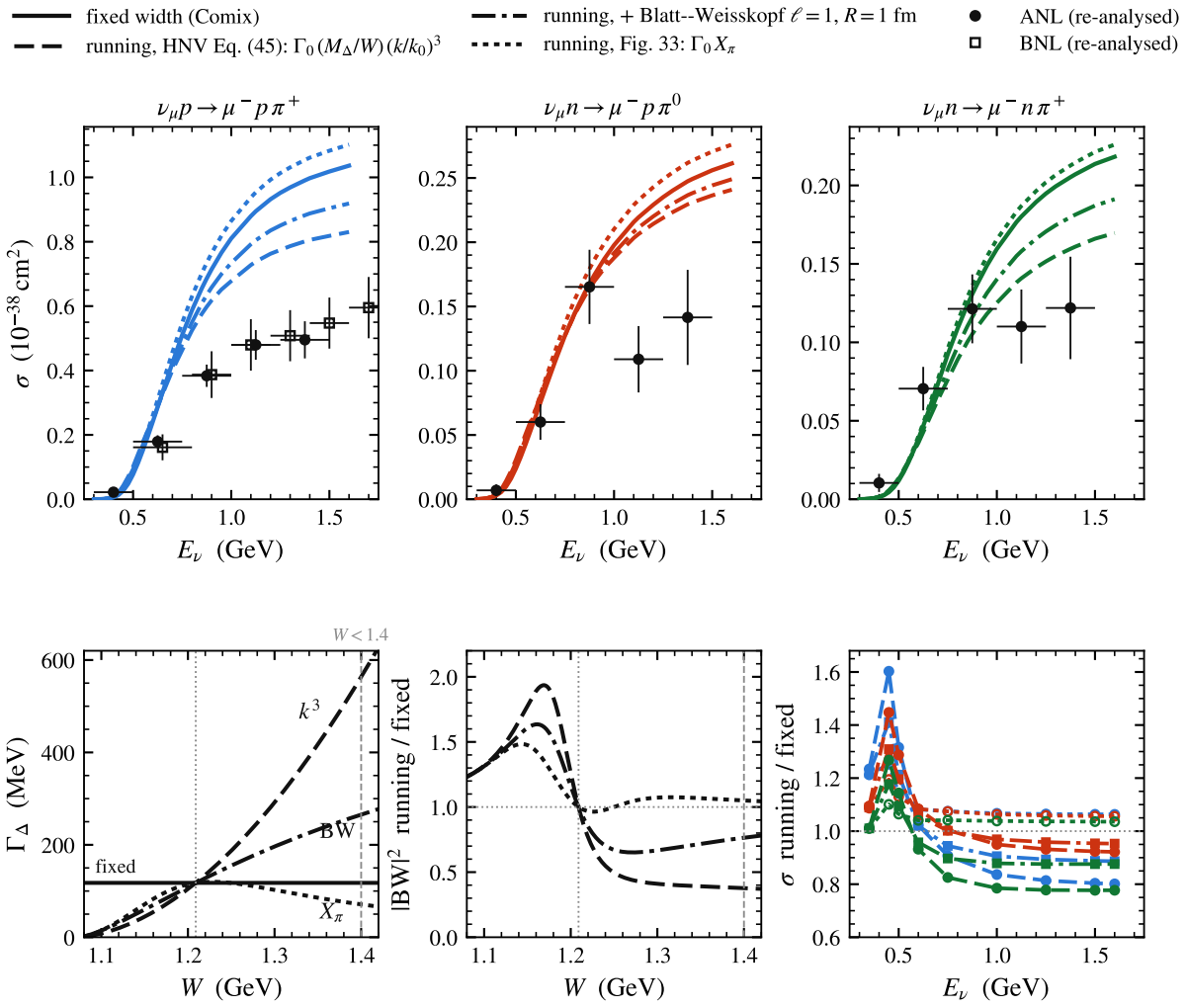


Figure 2: What an energy-dependent Δ width would do. **Top:** $\sigma(E_\nu)$ for the three charged-current single-pion channels with $W < 1.4$ GeV on both sides, against the re-analysed ANL and BNL data. Solid is the generator result at fixed width; the dashed families are that curve multiplied by the ratio $R(E_\nu)$ computed analytically. **Bottom left:** the width itself, all three prescriptions crossing at Γ_0 by construction. **Bottom centre:** the resulting change in the Breit–Wigner modulus squared, which enhances below the pole and suppresses above it and crosses unity exactly at M_Δ — the reason the effect changes sign. **Bottom right:** the integrated consequence, $R(E_\nu)$, for every channel and prescription. Dashed vertical lines mark the $W < 1.4$ GeV cut. See Sec. 5.